NEXUS 4

HOME SAFETY

IOT - BASED GAS LEAKAGE DETECTOR

Ishan Jiji George

Sherry John

Manuel James

Sona Salu

June 2025

INDEX

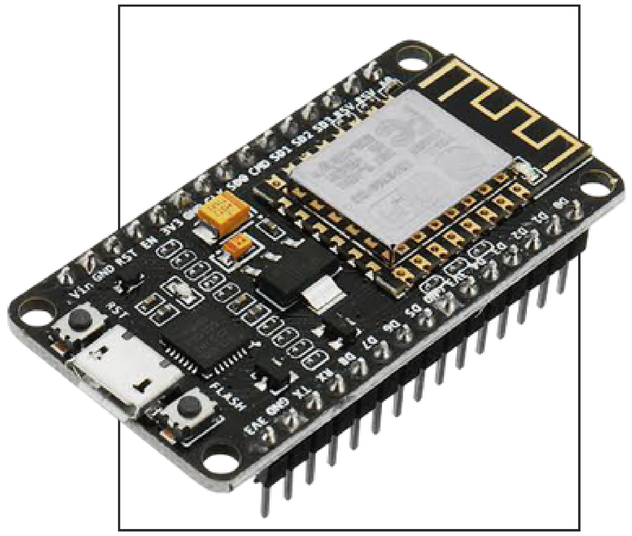
Introduction

- ESP8266: Controls system and connects to Wi-Fi.
- MQ6 Sensor: Detects flammable gases.
- DHT11 Sensor: Monitors temperature & humidity.
- Alerts: LED turns on instantly; buzzer after 2 seconds.
- Firebase: Real-time data upload for remote monitoring.

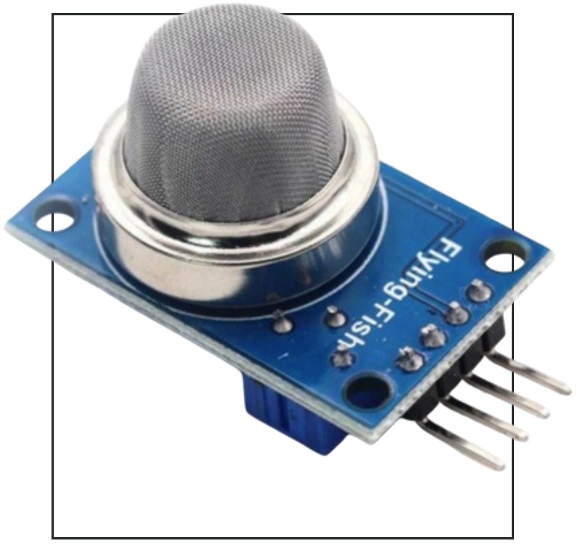
Project Overview

- Purpose: Detect gas leaks using ESP8266 and MQ6 sensor.
- Alert: LED turns on instantly; buzzer sounds after 2 seconds.
- Monitoring: DHT11 measures temperature/humidity; data sent to Firebase.
- Components: ESP8266, MQ6, DHT11, LED, buzzer, Firebase.

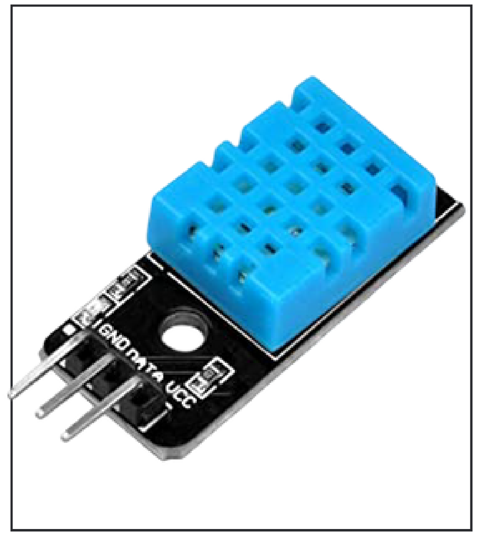
KEY COMPONENTS USED



ESP - 8266



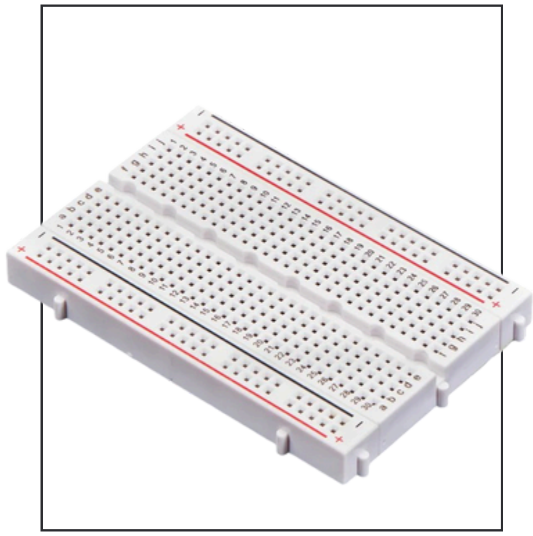
MQ-6 Gas Detector



DTH-11 Sensor

Jumper Wire

Buzzer



BreadBoard

LED

Implementation Plan

Hardware Setup

- ESP8266 NodeMCU for control & Wi-Fi
- MQ6 sensor to detect gas
- DHT11 for temperature & humidity
- LED and Buzzer as output indicators
- Basic breadboard or PCB wiring

Communication & Data Processing

- Connect ESP8266 to Wi-Fi
- Firebase used as real-time cloud DB
- Upload:
- Gas status
- Temperature & humidity
- Timestamp

Implementation Plan

Programming

- Arduino IDE with libraries: ESP8266WiFi, FirebaseESP8266, DHT
- Logic:
- Detect gas → LED ON
- After 2 sec → Buzzer ON
- Read DHT11
- Send all data to Firebase

Testing & Debugging

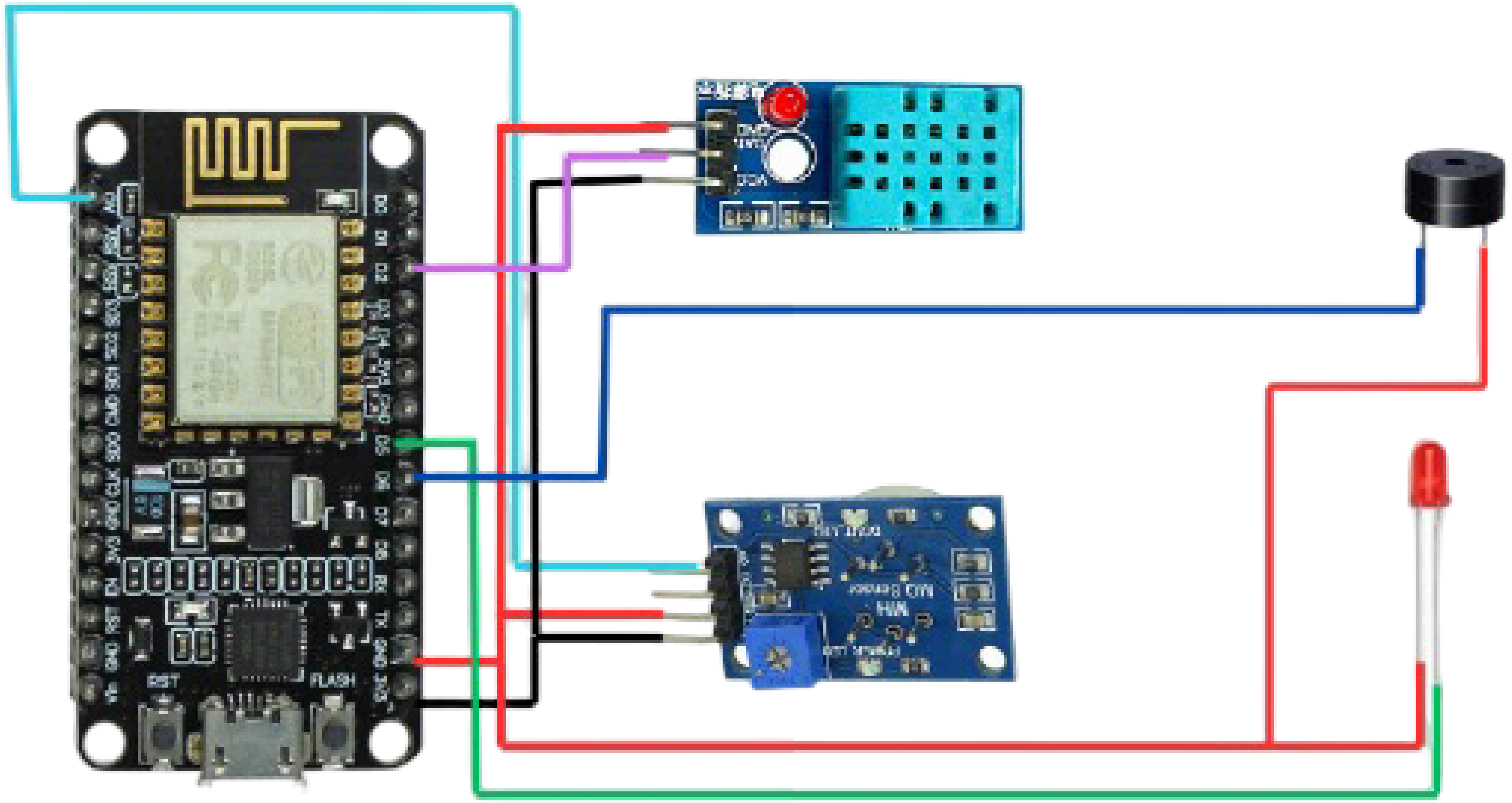
- Simulate gas leak using a lighter
- Check:
- LED and buzzer behavior
- Serial Monitor for live logs
- Firebase for real-time data update

Implementation Plan

Final Assembly & Packaging

- Assemble all components neatly on breadboard
- Use a protective outer casing with sensor opening
- Ensure airflow for MQ6 sensor

CIRCUIT DIAGRAM



EXPECTED IMPACT & BENEFITS

Enhanced Safety

Detects gas leaks instantly to help prevent accidents and health risks.

Timely Alerts

LED and buzzer provide immediate local warning upon gas detection.

Scalable Design

Can be upgraded with app alerts, smart valve control, or IoT integration.

Real-Time Monitoring

Sends live data to Firebase for remote monitoring from anywhere.

Cost-Effective

Built using affordable components, making it accessible for homes and small industries.

Easy Deployment

Compact design and breadboard setup make it easy to install and demonstrate in real environments.

FUTURE SCOPE

Mobile App Alerts

Integration with a mobile app to notify users instantly in case of gas leakage.

Automatic Gas Valve Shutoff

System upgrade to automatically close gas valves during leaks.

Battery Backup Support

Add rechargeable battery support for uninterrupted operation during power failure.

Voice Assistant Integration

Compatibility with Alexa/Google Assistant for smart home automation.

Conclusion

- The system effectively detects gas leaks using the MQ6 sensor.
- Immediate alerts are provided through LED and buzzer.
- Temperature and humidity data are recorded using the DHT11 sensor.
- Real-time data is successfully stored and monitored via Firebase.
- The project offers a reliable, low-cost safety solution for domestic and industrial use.